6. *(a)* The meaning of the word "latent" is "hidden". In terms of latent heat it suggests that the heat does something that is hidden.
The value of the specific latent heat of fusion for ice is 334 000 J/kg and the value of the specific latent heat of vaporisation for water is 2 260 000 J/kg.
Suggest why the word "latent" is used and account for the difference in the two values in terms of the behaviour of molecules. [6 QER]

Examiner only

(b) (i) A recycling company aims to melt 500 kg of aluminium in one hour. The aluminium, at an initial temperature of 20 °C is to be completely melted at 660 °C in a furnace of power 87 kW.

Use equations from page 2 and the information below to determine whether it is possible to meet the deadline. [6]

Specific heat capacity of aluminium, c = 900 J/kg °C
Melting point of aluminium = 660 °C
Specific latent heat of fusion of aluminium, L = 400 000 J/kg

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(ii) State why, in practice, it would take a lot longer than the time calculated above. [1]

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